Statistics 280 — Practice Midterm 2B — Solutions

Point values per part match the exam. Normal-table values follow the table at the back of the exam. R answers were run in R 4.3 (the ## lines are real output); any equivalent correct R command earns full credit.

Q.1 (30)

a (4 points)
Whatever the first student picks, the second matches with chance $1/10{,}000 = 0.0001$.

b (7 points)
$\frac{10000}{10000} \cdot \frac{9999}{10000} \cdots \frac{9971}{10000} = \prod_{i=0}^{29} \frac{10000-i}{10000} \approx 0.957$.

```
prod((10000 - 0:29) / 10000)
```

```
## [1] 0.9573915
```

c (6 points)
$1 - (9999/10000)^{29} \approx 0.0029$. The chance that some pair matches is $1 - 0.957 = 0.043$, about 15 times bigger. There are $\binom{30}{2} = 435$ pairs that could match, but only 29 of them involve you. This is the birthday problem's surprise (Lecture 10).

d (7 points)
Blanks: `0.5`, `n + 1`, `n`. The loop stops at the first n where P(all different) ≤ 0.5.

```
n = 1
while (prod((10000 - 0:(n - 1)) / 10000) > 0.5) {
  n = n + 1
}
print(n)
```

```
## [1] 119
```

e (6 points)
One six in 4 rolls: $1 - (5/6)^4 \approx 0.518$. One double-six in 24 rolls: $1 - (35/36)^{24} \approx 0.491$. The single six in 4 rolls is more likely. Both use the complement trick: P(at least one) = 1 − P(none).

Q.2 (30)

a (6 points)
$P(A) = 1/2$, $P(B) = 6/36 = 1/6$. A and B happen for (2,5), (4,3), (6,1), so $P(A \text{ and } B) = 3/36 = 1/12$. Since $1/12 = \frac{1}{2} \cdot \frac{1}{6}$, they **are independent**.

b (6 points)
$C = \{(2,6), (3,5), (4,4), (5,3), (6,2)\}$, so $P(C) = 5/36$. The first die is even in 3 of these, so

$P(A \text{ and } C) = 3/36$. But $P(A)P(C) = 5/72 \neq 6/72$, so they're **not independent**. Equivalently, $P(A|C) = 3/5 \neq 1/2$.

c (6 points)
They're mutually exclusive, since the sum can't be both 7 and 8. They're **not** independent: $P(B \text{ and } C) = 0$ but $P(B)P(C) > 0$. Knowing B happened tells you C did not.

d (5 points)
A and D concern different dice, so they're independent: $P(A \text{ or } D) = \frac{1}{2} + \frac{1}{6} - \frac{1}{2} \cdot \frac{1}{6} = \frac{7}{12} \approx 0.58$.

e (7 points)
$P(B \text{ and } D) = P((1,6)) = 1/36 = P(B)P(D)$, so they **are independent**. Given that the second die is 6, the sum is 7 exactly when the first die is 1, which has chance 1/6. That's the same as P(B). Whatever the second die shows, exactly one value of the first die makes the sum 7. (This isn't true for a sum of 8: if the second die is 1, no first die gives 8.)

Q.3 (40)

a (6 points)
Law of total probability: $P(T = 6) = \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{5}{36} = \frac{6}{72} + \frac{5}{72} = \frac{11}{72} \approx 0.15$.

b (6 points)
Bayes' rule: $P(H|T = 6) = \frac{(1/2)(1/6)}{11/72} = \frac{6}{11} \approx 0.55$.

c (4 points)
0. One die can't total 7 or more, so T ≥ 7 is only possible after tails.

d (6 points)
Heads: $P(T \leq 3|H) = 3/6$. Tails: totals 2 or 3 come from (1,1), (1,2), (2,1), so $P(T \leq 3|\text{tails}) = 3/36$. $P(H|T \leq 3) = \frac{(1/2)(1/2)}{(1/2)(1/2)+(1/2)(1/12)} = \frac{1/4}{7/24} = \frac{6}{7} \approx 0.86$.

e (6 points)
Prior odds of heads $= 1$. Data odds $= P(T = 6|H)/P(T = 6|\text{tails}) = (1/6)/(5/36) = 6/5$. Posterior odds $= 6/5$, so the probability is $\frac{6/5}{1+6/5} = 6/11$.

f (4 points)
If we repeat the experiment many times and keep only the runs where the total is 6, about 6 out of every 11 of those runs had heads.

g (8 points)
One solution, with a loop (vectorizing also works):

```
set.seed(3)
heads = c(); totals = c()
for (i in 1:100000) {
  h = sample(c(0, 1), 1)
  if (h == 1) {
    t = sample(1:6, 1)
  } else {
    t = sum(sample(1:6, 2, replace = TRUE))
  }
  heads = c(heads, h); totals = c(totals, t)
}
mean(heads[totals == 6])     # estimates 6/11 = 0.545
```

```
## [1] 0.5421766
```

Q.4 (30)

a (4 points)
$\binom{15}{5} = \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11}{5!} = 3003.$

b (6 points)
$\binom{9}{3}\binom{6}{2} / \binom{15}{5} = 84 \cdot 15 / 3003 = 1260 / 3003 \approx 0.42.$

c (6 points)
Ordered officers: $15 \cdot 14 \cdot 13 = 2730$. Then 2 members from the remaining 12: $\binom{12}{2} = 66$. Total $= 180{,}180$ ways. P(president and VP both seniors) $= \frac{9}{15} \cdot \frac{8}{14} = \frac{72}{210} \approx 0.34$. The treasurer and members don't matter.

d (6 points)
Choose which 6 of the 15 spots hold juniors: $\binom{15}{6} = 5005$ equally likely patterns. A block of 6 adjacent juniors can start at positions 1 through 10, so there are 10 patterns and the chance is $10 / 5005 \approx 0.002$.

e (8 points)
$P(\geq 4) = \frac{\binom{7}{4}\binom{8}{1} + \binom{7}{5}\binom{8}{0}}{\binom{15}{5}} = \frac{35 \cdot 8 + 21}{3003} = \frac{301}{3003} \approx 0.10.$

```
1 - phyper(3, 7, 8, 5)
```

```
## [1] 0.1002331
```

A 10% chance is unusual but not very unusual, so this alone isn't strong evidence that the committee was chosen with AP Statistics in mind. This is the reasoning of Fisher's exact test (Lecture 12).

Q.5 (40)

a (6 points)
$P(X=2, Y=2) = \frac{2}{4} \cdot \frac{1}{3} = \frac{1}{6}$. $P(Y=5) = \frac{1}{4}$: by symmetry the second ticket is equally likely to be any of the four tickets, just as the second card is an ace with chance 4/52.

b (6 points)
$E(X) = (1+2+2+5)/4 = 2.5$. $E(X^2) = (1+4+4+25)/4 = 8.5$, so $SE(X) = \sqrt{8.5 - 6.25} = 1.5$. **Yes**, they're the same: Y has the same distribution as X, since each of the four tickets is equally likely to be drawn second.

c (8 points)
Over the 12 equally likely ordered pairs of different tickets, the sum of the products is $(\sum x_i)^2 - \sum x_i^2 = 10^2 - 34 = 66$. So $E(XY) = 66/12 = 5.5$, and $\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 5.5 - 6.25 = -0.75$.

d (8 points)
$\text{Var}(X+Y) = 2.25 + 2.25 + 2(-0.75) = 3$, so $SE(X+Y) = \sqrt{3} \approx 1.73$. $\text{Var}(X-Y) = 2.25 + 2.25 - 2(-0.75) = 6$, so $SE(X-Y) = \sqrt{6} \approx 2.45$. With replacement, X and Y are independent: $SE(X+Y) = \sqrt{4.5} \approx 2.12$. Without replacement the covariance is negative: a big first draw leaves smaller tickets for the second, so the sum varies less.

e (8 points)

```
set.seed(5)
N = 100000
x = numeric(N); y = numeric(N)
for (i in 1:N) {
  t = sample(c(1, 2, 2, 5), 2)
  x[i] = t[1]; y[i] = t[2]
}
cov(x, y) * (N - 1) / N            # about -0.75
sd(x + y) * sqrt((N - 1) / N)      # about 1.73
```

```
## [1] -0.7552166
## [1] 1.730855
```

f (4 points)
No. Review 13 Q26: X uniform on {−4, 0, 4} has $\text{Cov}(X, X^2) = 0$, yet X^2 is completely determined by X. Lecture 13's tickets (0,0), (1,1), (2,1), (3,0) are another example.

Q.6 (30)

a (6 points)
38 tickets: 18 marked +1 and 20 marked −1, drawn 100 times with replacement. Average $= (18 - 20)/38 \approx -\$0.053$. SD $= (b - s)\sqrt{p_b p_s} = 2\sqrt{\frac{18}{38} \cdot \frac{20}{38}} \approx \$1.00$.

b (6 points)
EV $= 100 \times (-0.053) \approx -\$5.26$. SE $= \sqrt{100} \times 1.00 \approx \$9.99$.

c (8 points)
Let W be the number of wins. Net $= W - (100 - W) = 2W - 100 > 0$ exactly when $W \geq 51$. W has EV $100 \cdot 18/38 = 47.37$ and SE $\sqrt{100 \cdot \frac{18}{38} \cdot \frac{20}{38}} = 4.99$. With the continuity correction: $z = (50.5 - 47.37)/4.99 = 0.63$. The area within 0.63 is about 47.1%, so the chance is about $(100 - 47.1)/2 \approx \mathbf{26\%}$.

```
1 - pbinom(50, 100, 18/38)
```

```
## [1] 0.2650235
```

d (5 points)
W has EV 4736.8 and SE $\sqrt{10000 \cdot \frac{18}{38} \cdot \frac{20}{38}} = 49.9$. Ahead means $W \geq 5001$: $z = (5000.5 - 4736.8)/49.9 \approx 5.3$. That's beyond the table, so the chance is essentially 0 (less than 0.001%).

e (5 points)
The expected loss grows in proportion to the number of bets, n: −$5.26 per 100 bets. The SE grows only like $\sqrt{n}$. In percentage terms the chance error shrinks (law of averages), so over many bets the casino's edge almost surely wins.

Q.7 (15 extra credit)

Average $\mu = bp_b + sp_s$, and the average of the squares is $b^2p_b + s^2p_s$. By the shortcut formula,

$$SD^2 = b^2p_b + s^2p_s - (bp_b + sp_s)^2 = b^2p_b(1 - p_b) + s^2p_s(1 - p_s) - 2bs\,p_bp_s.$$

Since $1 - p_b = p_s$ and $1 - p_s = p_b$, this is $p_bp_s(b^2 + s^2 - 2bs) = p_bp_s(b - s)^2$. Taking square roots gives $SD = (b - s)\sqrt{p_bp_s}$.